

Other parameters such as depletion-region capacitance and breakdown voltage become important only under certain operating conditions and will be considered only where appropriate.

SECTION 3 CHECKUP

1. Draw a rectifier diode symbol and label the terminals.
2. For a normal diode, the forward resistance is quite low and the reverse resistance is very high. (True or False)
3. An open switch ideally represents a _____-biased diode. A closed switch ideally represents a _____-biased diode.
4. How does the practical model of a diode differ from the ideal model?



4 DIODE RECTIFIERS

Because of their ability to conduct current in one direction and block current in the other direction, diodes are used in circuits called **rectifiers** that convert ac voltage into dc voltage. Rectifiers are found in all dc power supplies that operate from an ac voltage source.

After completing this section, you should be able to

- ♦ **Analyze the operation of a half-wave rectifier and a full-wave rectifier**
 - ♦ Describe a basic dc power supply
 - ♦ Explain the processes of half-wave and full-wave rectification
 - ♦ Describe how the diode functions in a rectifier
 - ♦ Determine the average value of a half-wave rectified voltage
 - ♦ Determine the average value of a full-wave rectified voltage
 - ♦ Define *peak inverse voltage (PIV)*

The Half-Wave Rectifier

Figure 22 illustrates the process called *half-wave rectification*. In part (a), a diode is connected to an ac source that provides the input voltage, V_{in} , and to a load resistor, R_L , forming a **half-wave rectifier**. Keep in mind that all ground symbols represent the same point electrically. Let's examine what happens during one cycle of the input voltage using the ideal model for the diode. When the sinusoidal input voltage goes positive, the diode is forward-biased and conducts current through the load resistor, as shown in part (b). The current produces an output voltage across the load, which has the same shape as the positive half-cycle of the input voltage.

When the input voltage goes negative during the second half of its cycle, the diode is reverse-biased. There is no current, so the voltage across the load resistor is zero, as shown in Figure 22(c). The net result is that only the positive half-cycles of the ac input voltage appear across the load. Since the output does not change polarity, it is a pulsating dc voltage, as shown in part (d).

Average Value of the Half-Wave Rectified Output Voltage The average value of a half-wave rectified output voltage is the value you would measure on a dc voltmeter. It can be calculated with the following equation where $V_{p(out)}$ is the peak value of the half-wave rectified output voltage:

$$V_{AVG} = \frac{V_{p(out)}}{\pi}$$

Equation 1

► **FIGURE 22**

Operation of a half-wave rectifier.

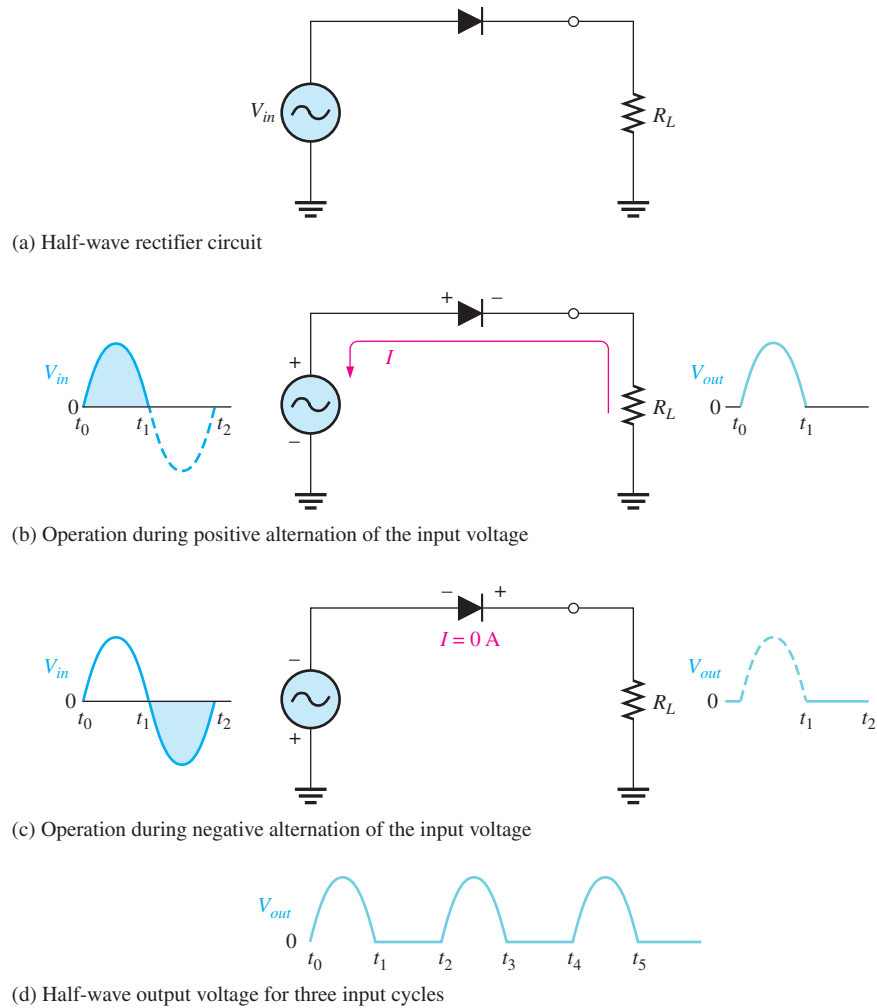
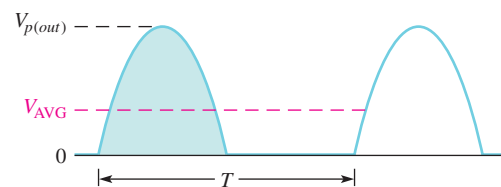


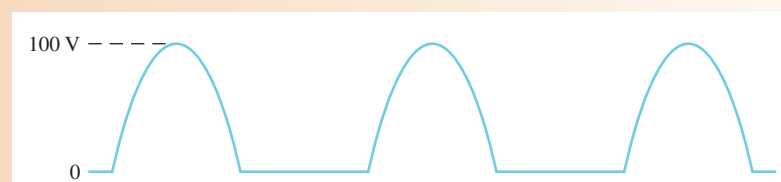
Figure 23 shows the half-wave voltage with its average value indicated by the red dashed line.

► **FIGURE 23**

Average value of the half-wave rectified signal.

**EXAMPLE 1**

What is the average (dc) value of the half-wave rectified output voltage waveform in Figure 24?

► **FIGURE 24**

Solution

$$V_{\text{AVG}} = \frac{V_{p(\text{out})}}{\pi} = \frac{100 \text{ V}}{\pi} = \mathbf{31.8 \text{ V}}$$

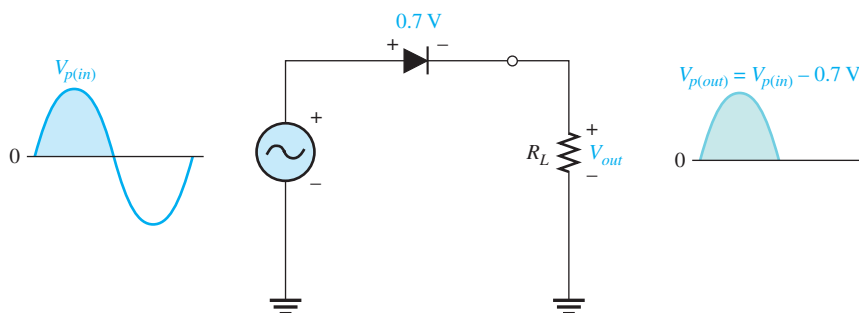
Related Problem*

Determine the average value of the half-wave rectified output voltage if its peak amplitude is 12 V.

*Answers are at the end of the chapter.

Effect of Diode Barrier Potential on Half-Wave Rectifier Output Voltage In the previous discussion, the diode was considered ideal. When the diode barrier potential is taken into account as in the practical model discussed in Section 3, here is what happens: During the positive half-cycle, the input voltage must overcome the barrier potential before the diode becomes forward-biased. This results in a half-wave output voltage with a peak value that is 0.7 V less than the peak value of the input voltage, as shown in Figure 25. The expression for peak output voltage is

$$V_{p(\text{out})} = V_{p(\text{in})} - 0.7 \text{ V}$$

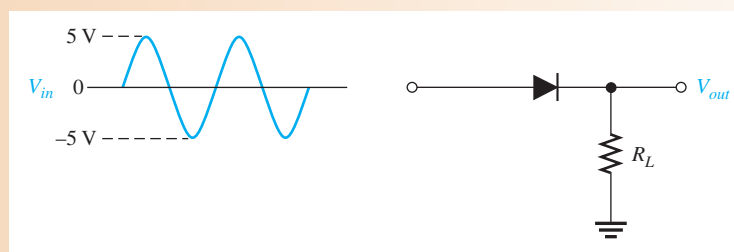
Equation 2**FIGURE 25**

Effect of barrier potential on half-wave rectified output voltage.

When you work with diode circuits, it is often practical to neglect the effect of barrier potential when the peak value of the applied voltage is much greater (at least ten times) than the barrier potential.

EXAMPLE 2

Determine the peak output voltage and the average value of the output voltage of the rectifier in Figure 26 for the indicated input voltage.

FIGURE 26**Solution**

$$V_{p(\text{out})} = V_{p(\text{in})} - 0.7 \text{ V} = 5 \text{ V} - 0.7 \text{ V} = \mathbf{4.3 \text{ V}}$$

$$V_{\text{AVG}} = \frac{V_{p(\text{out})}}{\pi} = \frac{4.3 \text{ V}}{\pi} = \mathbf{1.37 \text{ V}}$$

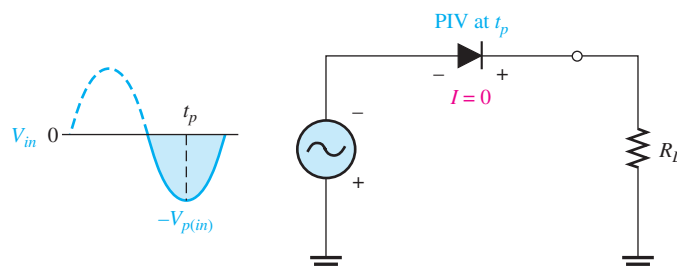
Related Problem

Determine the peak output voltage for the rectifier in Figure 26 if the peak input is 3 V.

Peak Inverse Voltage (PIV) The maximum value of reverse voltage, sometimes designated as **PIV** (peak inverse voltage), occurs at the peak of each negative alternation of the input cycle when the diode is reverse-biased. This condition is illustrated in Figure 27. The PIV equals the peak value of the input voltage, and the diode must be capable of withstanding this amount of repetitive reverse voltage.

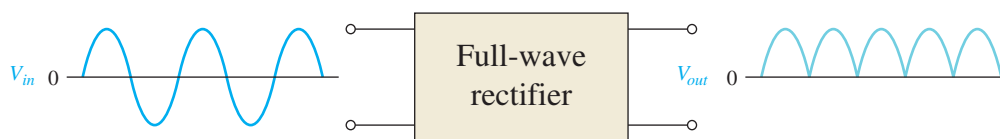
► **FIGURE 27**

The PIV occurs at the peak of each half-cycle of the input voltage when the diode is reverse-biased. In this circuit, the PIV occurs at the time (t_p) of the peak of each negative half-cycle.



The Full-Wave Rectifier

The difference between full-wave and half-wave rectification is that a **full-wave rectifier** allows unidirectional current to the load during the entire input cycle and the half-wave rectifier allows this only during one-half of the cycle. The result of full-wave rectification is a dc output voltage that pulsates every half-cycle of the input, as shown in Figure 28.



▲ **FIGURE 28**

Full-wave rectification.

The average value for a full-wave rectified output voltage is twice that of the half-wave rectified output voltage, expressed as follows:

Equation 3

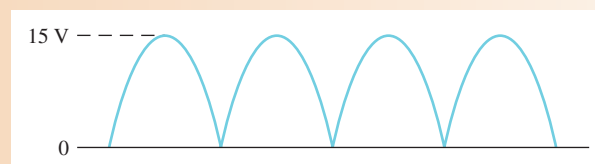
$$V_{\text{AVG}} = \frac{2V_{p(\text{out})}}{\pi}$$

Since $2/\pi = 0.637$, you can calculate V_{AVG} as $0.637V_{p(\text{out})}$.

EXAMPLE 3

Find the average value of the full-wave rectified output voltage in Figure 29.

► **FIGURE 29**



Solution

$$V_{\text{AVG}} = \frac{V_{p(\text{out})}}{\pi} = \frac{2(15 \text{ V})}{\pi} = 9.55 \text{ V}$$

Related Problem Find V_{AVG} for the full-wave rectified voltage if its peak is 155 V.

Center-Tapped Full-Wave Rectifier

The center-tapped (CT) full-wave rectifier uses two diodes connected to the secondary of a center-tapped transformer, as shown in Figure 30. The input signal is coupled through the transformer to the secondary. Half of the secondary voltage appears between the center tap and each end of the secondary winding as shown.

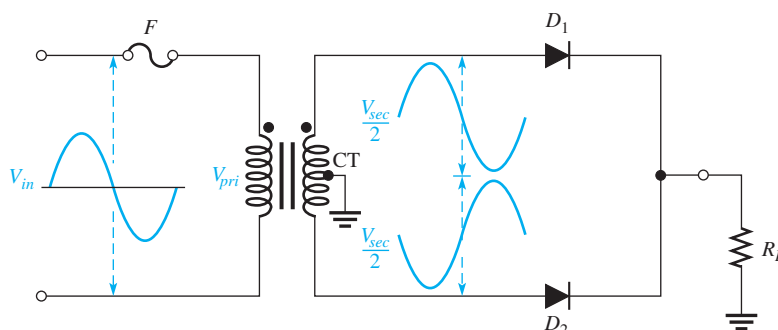
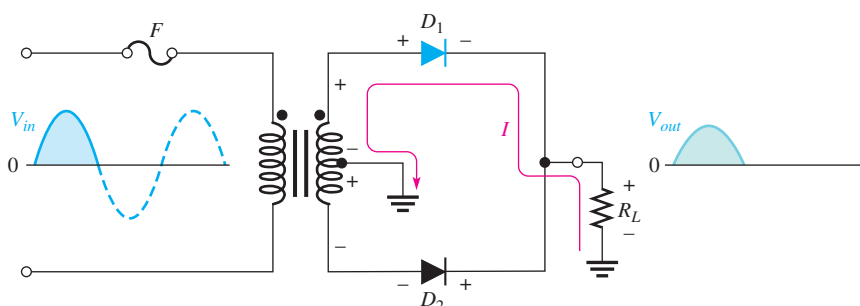


FIGURE 30

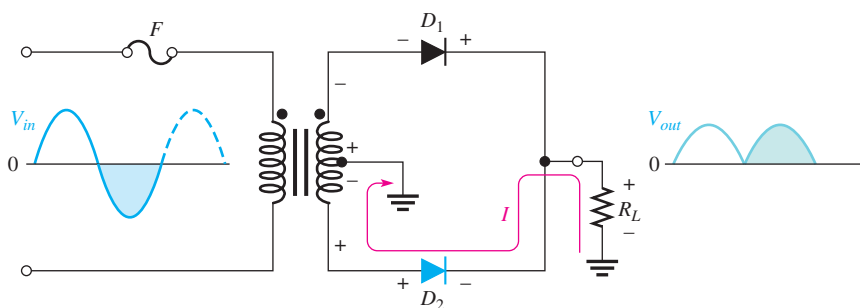
A center-tapped full-wave rectifier.

$$V_{pri} = V_{in}$$

For a positive half-cycle of the input voltage, the polarities of the secondary voltages are as shown in Figure 31(a). This condition forward-biases the upper diode D_1 and reverse-biases the lower diode D_2 . The current path is through D_1 and the load resistor R_L , as indicated. For a negative half-cycle of the input voltage, the voltage polarities on the secondary are as shown in Figure 31(b). This condition reverse-biases D_1 and forward-biases D_2 . The current path is through D_2 and R_L , as indicated. Because the current during both the positive and the negative portions of the input cycle is in the same direction through the load, the output voltage developed across the load is a full-wave rectified dc voltage.



(a) During positive half-cycles, D_1 is forward-biased and D_2 is reverse-biased.



(b) During negative half-cycles, D_2 is forward-biased and D_1 is reverse-biased.

FIGURE 31

Basic operation of a center-tapped full-wave rectifier. Note that the current through the load resistor is in the same direction during the entire input cycle.

HANDS ON TIP

A diode is sometimes used to isolate two power sources such as in the case of a battery backup for a

computer system. The power from the power supply is connected through a series diode to the power input point on the system. A battery backup system is connected to the same point through its own series diode. The battery backup is set for a slightly lower voltage so that it does not discharge the battery during normal operation. When a power failure occurs, the battery backup system takes over and the diodes prevent the two power sources from affecting each other.